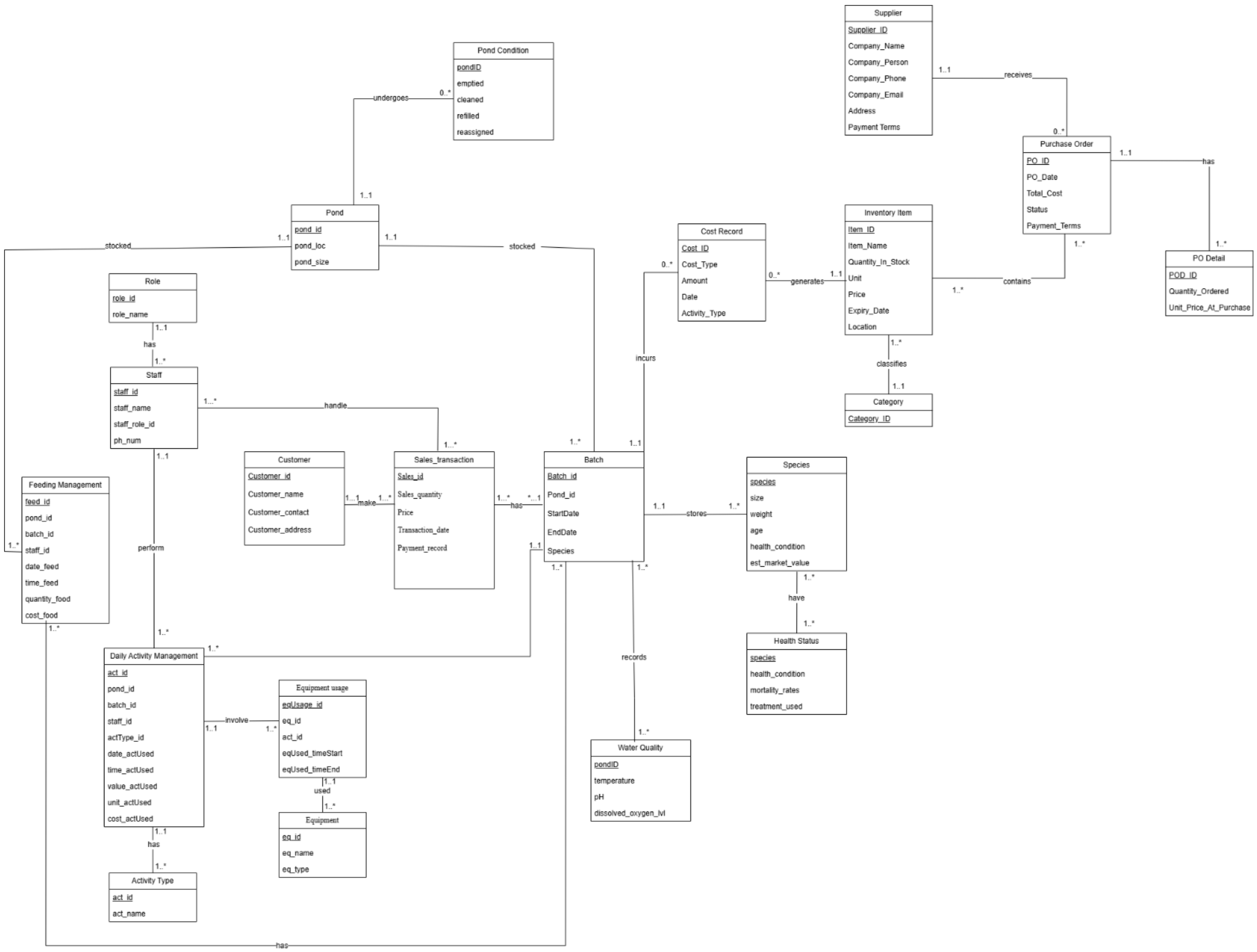
Table of content

1.0 ERD Diagram



2.0 Remove features not compatible with the relational model

2.1 Remove *:* Binary Relationship Types

a)

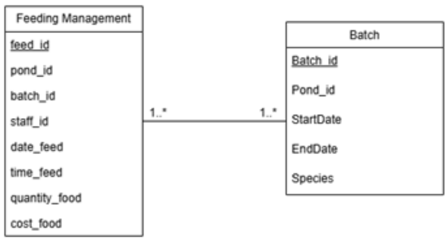


Figure 2.1.1(Before)

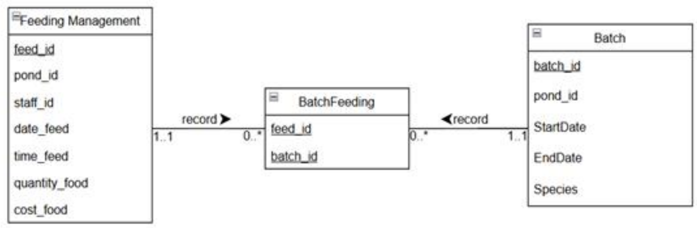


Figure2.1.2(After)

To remove the many to many relation, a junction entity called batch feeding is formed between them. The junction entity BatchFeeding uses a composite primary key consisting of both feed_id and batch_id. This ensures that each unique combination of a feeding event and a batch is recorded only once, maintaining data integrity while resolving the many-to-many relationship into two one-to-many relationships.While feed_id is chosen as a foreign key referencing Feeding Management and batch_id is chosen foreign key referencing Batch.

b)

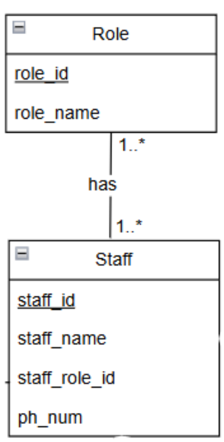


Figure 2.1.3(Before)

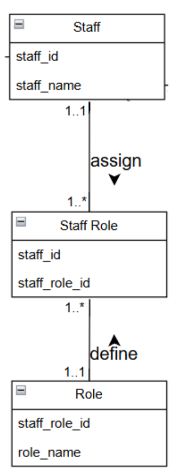


Figure 2.1.4(After)

To remove the many to many relation, a junction entity called Staff Assignments is formed between them. The junction entity Staff Assignments uses a composite primary key consisting of both staff_id and staff_role_id. The new relationships are both one-to-many (1 to 0..*), which are clean and implementable relationships.For example, one staff can have many staff role entries,while one role can have many staff role entries.

c)

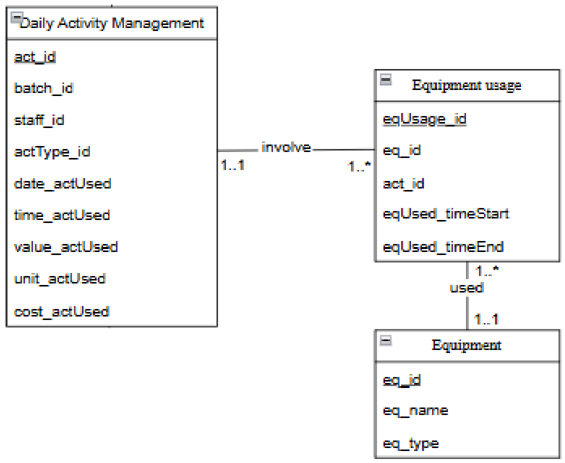


Figure 2.1.5

From 1.0 ERD, a junction entity called equiment usage is added between Daily Activity Managenent and Equipment. The eqUsage_id is chosen to be primary key, while act_id is chosen as a foreign key referencing Daily Activity Managenent and eq_id is chosen foreign key referencing Equipment.

d)

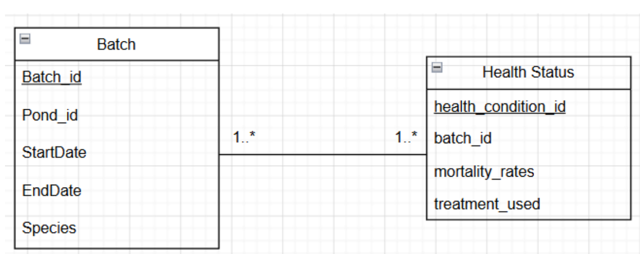


Figure 2.1.6 (Before)

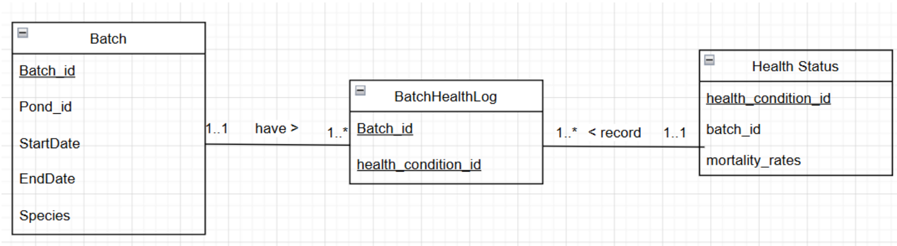


Figure 2.1.7 (After)

The many to many (*:*) binary relationship between the Batch and the Health status entities was resolved by introducing a junction entity named as Batch_Health_Log. This is because a single fish batch can undergo multiple health evaluations over its lifecycle while a specific health status can apply to various batches simultaneously by decomposing this relationship into two one-to-many links. So, Batch_id is chosen as a foreign key referencing Batch and batch_condition_id is chosen foreign key referencing Health Status. Thus, this structure not only satisfies the First Normal Form (1NF) by ensuring atonicity but also supports historical tracking of health changes over time.

e)

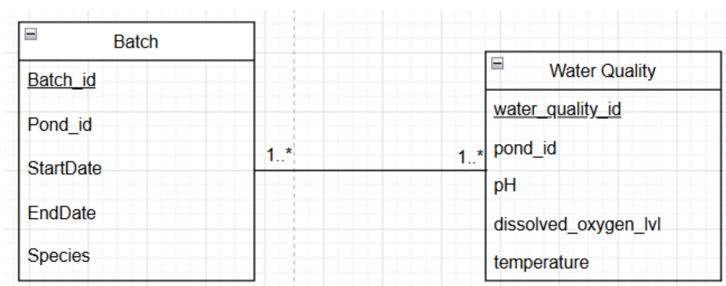


Figure 2.1.8 (Before)

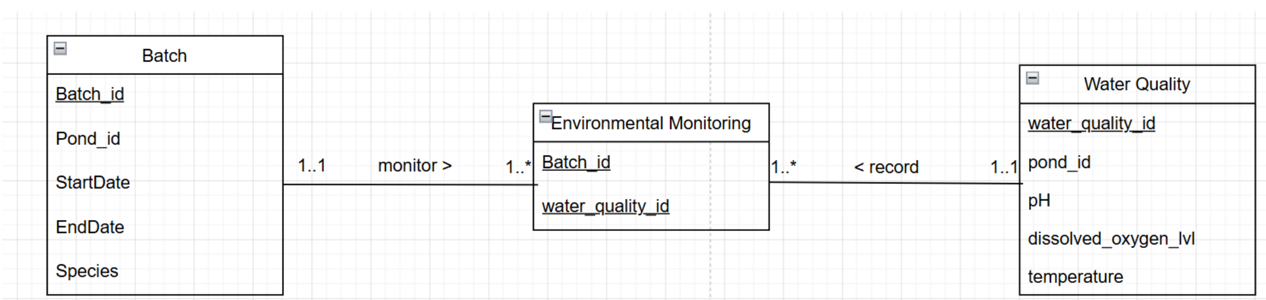


Figure 2.1.9 (After)

A junction entity was again created as Environmental Monitoring between the Batch and the Water Quality entities because a single batch of fish is subjected to numerous water quality readings throughout its production cycle while a single set of environmental data such as temperature, pH, and dissolved oxygen level can technically impact the health profile of the batch that is currently occupying the pond. Therefore, the relationship is broken down from many-to-many relationships into two one-to-many relationships in order to record the water readings to a particular batch in a more precise way without causing data redundancy. Batch_id is chosen as a foreign key referencing to Batch and water_quality_id is chosen foreign key referencing Water Quality..

f)

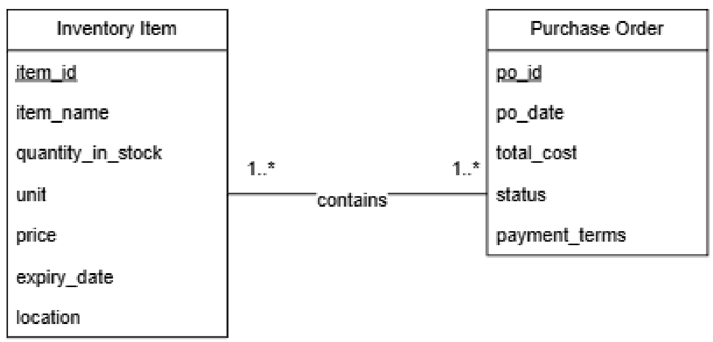


Figure 2.1.10 (before)

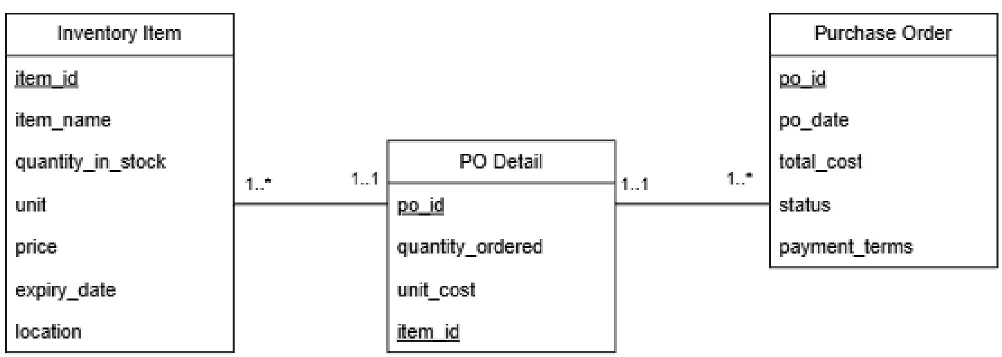


Figure 2.1.11 (after)

The many to many binary relationship between the Purchase Order and Inventory Item entities was resolved through the introduction of an associative entity named PO_Detail. This is because one purchase order may contain multiple inventory items, while one inventory item may appear in many purchase orders over time. By converting this relationship into two one to many relationships, data redundancy is reduced and transaction records can be managed efficiently. Po_id is used as a foreign key referencing Purchase Order, while item_id is used as a foreign key referencing Inventory Item.

2.2 Remove Multi-Valued Attributes

a)

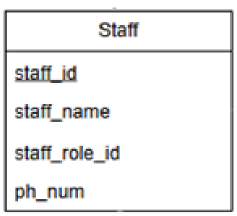


Figure 2.2.1(Before)

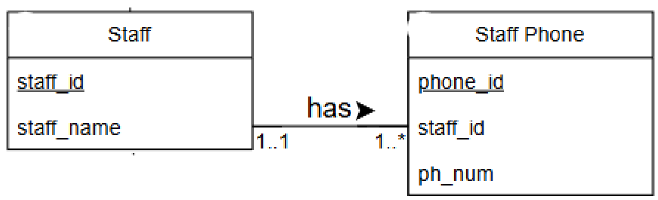


Figure 2.2.2(After)

In this figure 2.2.1 , ph_num is likely a multi-valued attribute, as an individual often has multiple contact numbers such as mobile, home and office. Thus, an entity Staff Phone is created and phone_id act as a primary key while staff_id act as a foreign key.

b)

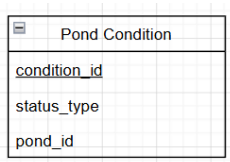


Figure 2.2.3 (Before)

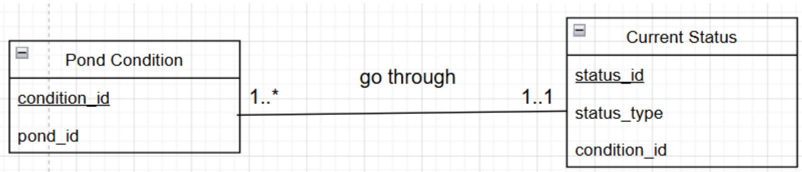


Figure 2.2.4 (After)

The attribute such as status_type was removed from the Pond Condition entity to be broken down into a new entity called Current Status so this transition resolves the restriction from First Normal Form (1NF) to make every attribute only have one value. Hence, condition_id will become the foreign key and status_ID as the primary key in the Current Status entity.

c)

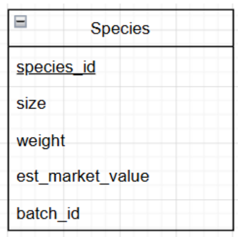


Figure 2.2.5 (Before)

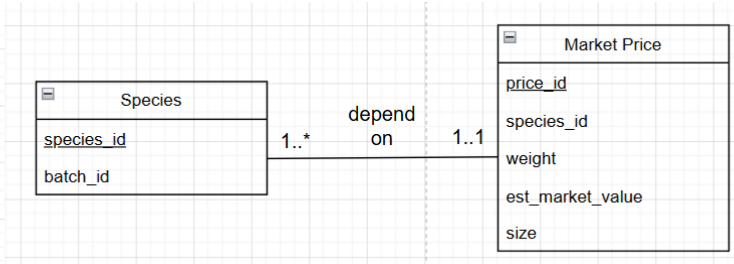


Figure 2.2.6 (After)

The estimated market price of the fish is expected to depend on the size, weight, age of the fish. Therefore, the attributes are placed in a new entity called Market Price in order to match the third normal form of Normalization. Species_id will act as the foreign key and price_id act as the primary key in the new entity.

3.0 Derive Relations For Local Logical Data Model

3.1 Strong Entity Types

A Strong Entity is independent and has its own unique primary key which means that it doesn't rely on another entity to exist.

Staff (staff_id, staff_name) Primary key staff_id	Supplier (supplier_id, company_name, company_person, company_phone, company_email, address, payment_terms) Primary key supplier_id
Role (staff_role_id, role_name) Primary key staff_role_id	Category (category_id, category_name) Primary key category_id
Equipment (eq_id, eq_name, eq_type) Primary key eq_id	Inventory Item (item_id, item_name, quantity_in_stock, unit, price, expiry_date, location) Primary key item_id

Activity Type (actType_id, act_name) Primary key actType_id	Purchase Order (po_id, po_date, total_cost, status, payment_terms) Primary key po_id
Pond (pond_id, pond_loc, pond_size) Primary key pond_id	Cost Record (cost_id, cost_type, amount, date, activity_type) Primary key cost_id
Species (Species_id, size, weight, age, health_condition, est_market_value) Primary key species_id	Sales_transaction (Sales_id, Customer_id, Sales_quantity, Price, Transaction_date ,Payment_record) Primary key Sales_id
Customer (Customer_id ,Customer_name, Customer_contact, Customer_address) Primary key Customer_id	

3.2 Weak Entity Types

A Weak Entity cannot exist without its parent entity. It often represents an event or a connection between two strong entities.

Staff Phone (phone_id, staff_id, ph_num) Primary key phone_id	PO Detail (po_id, quantity_ordered, unit_cost, item_id) Primary key po_id, item_id
Staff Role(staff_id, staff_role_id, description) Primary key staff_id, staff_role_id	Batch (batch_id, pond_id, StartDate, EndDate, Species) Primary key batch_id
Daily Activity Management (act_id, batch_id, staff_id, actType_id, date_actUsed, time_actUsed, value_actUsed, unit_actUsed, cost_actUsed) Primary key act_id	Pond Condition (condition_id, emptied, cleaned, refilled, reassigned, pond_id) Primary key condition_id

Equipment usage (eqUsage_id, eq_id, act_id, eqUsed_timeStart, eqUsed_timeEnd) Primary key eqUsage_id	Water Quality (water_quality_id, pond_id, pH, dissolved_oxygen_lvl, temperature) Primary key water_quality_id
Feeding management (feed_id,pond_id,staff_id ,date_feed, time_feed,quantity_food,cost_food) Primary key feed_id	Health Status (health_condition_id, batch_id, mortality_rates, treatment_used) Primary key health_condition_id
Batch Feeding(feed_id,batch_id) Primary key feed_id,batch_id	

3.3 One To Many(1..*)Binary Relation Types

Post pond_id into Batch

Pond (pond_id,pond_loc,pond_size)

Primary key pond_id

Batch (batch_id, pond_id, StartDate, EndDate, Species)

Primary key batch_id

Foreign key pond_id references Pond(pond_id)

Post **pond_id** into **Pond Condition**

Pond (pond_id, pond_loc, pond_size)
Primary Key (pond_id)

Pond Condition (condition_id)
Primary Key condition_id
Foreign Key pond_id **references** Pond(pond_id)

Post **batch_id** into **Species**

Batch (Batch_id, Pond_id, StartDate, EndDate, Species)
Primary Key Batch_id

Species (species_id, batch_id)
Primary Key species_id, batch_id
Foreign Key species_id **references** Species(species_id)
Foreign Key batch_id **references** Batch(batch_id)

Post **Customer_id** into **Sales_transaction** to model the 1:* makes/has relationship

Customer (Customer_id, Customer_name, Customer_contact ,Customer_address)
Primary key customer_id

Sales_transaction (Sales_id, Customer_id, Sales_quantity, Price, Transaction_date ,Payment_record)

Primary key Sales_id
Foreign key Customer_id **references** Customer(Customer_id)

Post supplier_id into Purchase Order

Supplier (supplier_id, company_name, company_person, company_phone, company_email, address, payment_terms)
Primary key supplier_id

Purchase Order (po_id, po_date, total_cost, status, payment_terms)
Primary key po_id
Foreign key supplier_id reference Supplier(supplier_id)

Post supplier_id into Inventory Item

Supplier (supplier_id, company_name, company_person, company_phone, company_email, address, payment_terms)
Primary key supplier_id

Inventory Item (item_id, item_name, quantity_in_stock, unit, price, expiry_date, location)
Primary key item_id
Foreign key supplier_id references Supplier(supplier_id)

Post category_id into Inventory Item

Category (category_id, category_name)
Primary key category_id

Inventory Item (item_id, item_name, quantity_in_stock, unit, price, expiry_date, location)
Primary key item_id
Foreign key category_id references Category(category_id)

Post po_id into Cost Record

Purchase Order (po_id, po_date, total_cost, status, payment_terms)
Primary key po_id

Cost Record (cost_id, cost_type, amount, date, activity_type)
Primary key cost_id
Foreign key po_id references Purchase Order(po_id)

Post batch_id into Cost Record

Batch (batch_id, pong_id, StartDate, EndDate, Species)
Primary key batch_id

Cost Record (cost_id, cost_type, amount, date, activity_type)
Primary key cost_id
Foreign key batch_id references Batch (batch_id)

3.4 ManyTo Many(*.. *)Binary Relation Types

Staff (staff_id,staff_name)

Primary key staff_id

Role (staff_role_id,role_name)

Primary key staff_role_id

Post staff_id into Staff Role

Post staff_role_id into Staff Role

Staff Role (staff_id, staff_role_id)

Primary key staff_id， staff_role_id

Foreign key staff_id references Staff(staff_id)

Foreign key staff_role_id references Role(staff_role_id)

Feeding management (feed_id, pond_id, staff_id ,date_feed, time_feed,quantity_food,cost_food)

Primary key feed_id

Foreign key pond_id references Pond(pond_id)

Foreign key staff_id references Staff(staff_id)

Batch (batch_id, pond_id, StartDate, EndDate, Species)

Primary key batch_id

Foreign key pond_id references Pond(pond_id)

Post feed_id into Batch Feeding

Post batch_id into Batch Feeding

Batch Feeding(feed_id,batch_id)

Primary key feed_id,batch_id

Foreign key feed_id references Feeding management(feed_id)

Foreign key batch_id references Batch (batch_id)

Post eq_id into Equipment usage

Post act_id into Equipment usage

Equipment(eq_id,eq_name,eq_type)

Primary key eq_id

Equipment usage(eqUsage_id, eq_id,act id,eqUsed_timeStart,eqUsed_timeEnd)

Primary key eqUsage_id

Foreign key eq_id references Equiment(eq_id)

Foreign key act_id references Daily Activity Management(act_id)

Daily Activity Management (act_id, batch_id, staff_id, actType_id,date_actUsed, time_actUsed, value_actUsed, unit_actUsed, cost_actUsed)

Primary key act_id

Foreign key actType_id references Activity Type(actType_id)

Foreign key batch_id references Batch(batch_id)

Foreign key staff_id references Staff(staff_id)

Batch (batch_id, Pond_id, StartDate, EndDate, Species)
Primary key Batch_id
Foreign key Pond_id **references** Pond(Pond_id)

Health Status (heatlh_condition_id, batch_id, mortality_rates)
Primary key health_condition_id
Foreign key Batch_id **references** Batch(Batch_id)

Post **Batch_id** into **BatchHealthLog**

Post **health_condition_id** into **BatchHealthLog**

BatchHealthLog (Batch_id, health_condition_id)
Primary key Batch_id, health_condition_id
Foreign key Batch_id **references** Batch(Batch_id)
Foreign key health_condition_id **references** Health Status (health_condition_id)

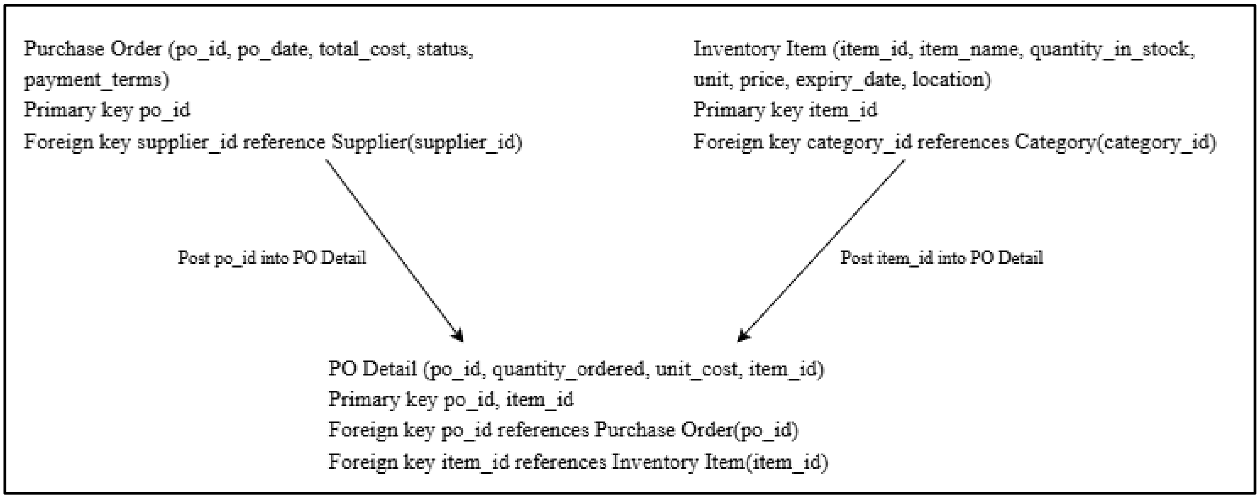
Batch (Batch_id, Pond_id, StartDate, EndDate, Species)
Primary Key Batch_id
Foreign Key Pond_id **references** Pond(Pond_id)

Water Quality (water_quality_id, pond_id, pH, dissolved_oxygen_lvl, temperature)
Primary Key water_quality_id
Foreign Key Pond_id **references** Pond(Pond_id)

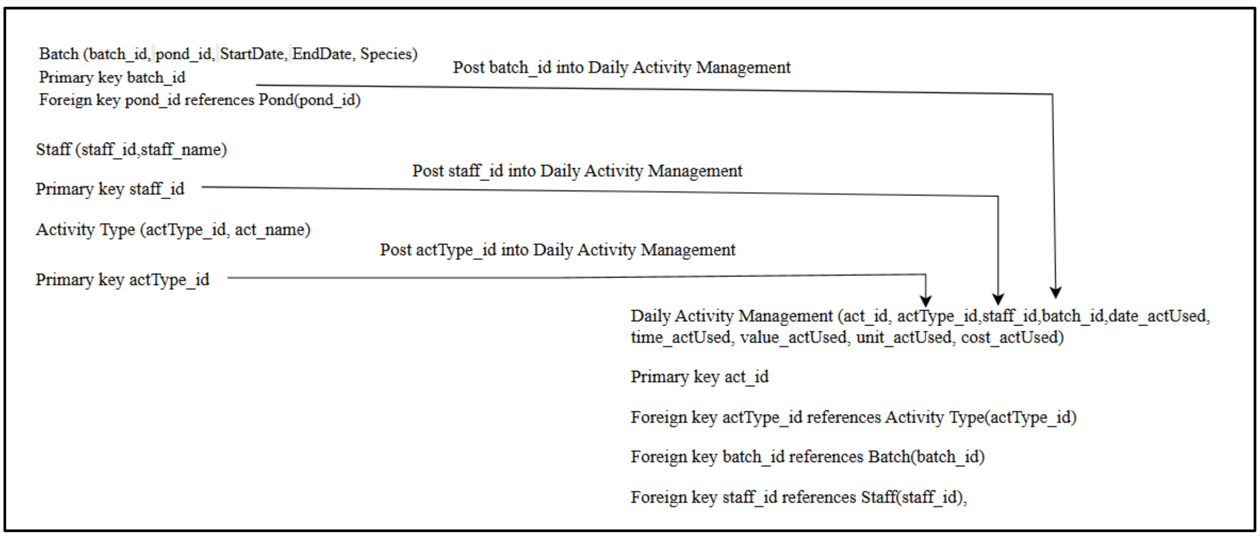
Post **Batch_id** into **Environmental Monitoring**

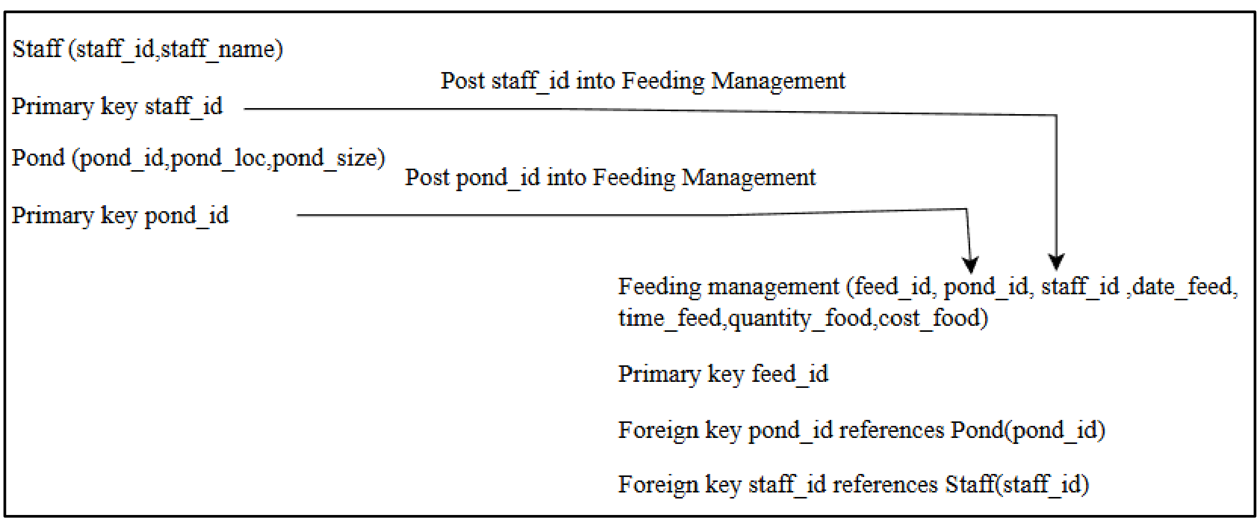
Post **water_quality_id** into **Environmental Monitoring**

Environmental Monitoring (Batch_id, Pond_id, StartDate, EndDate, Species)
Primary Key Batch_id, water_quality_id
Foreign Key Batch_id **references** Batch(Batch_id)
Foreign Key water_quality_id **references** Water Quality(water_quality_id)

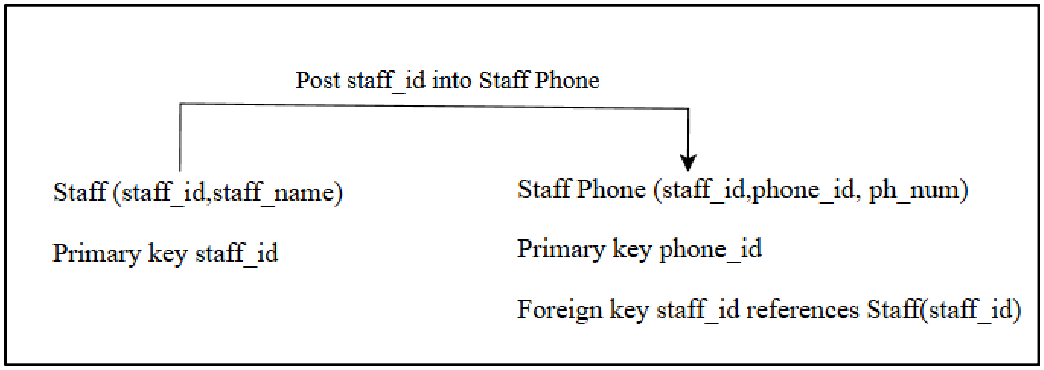


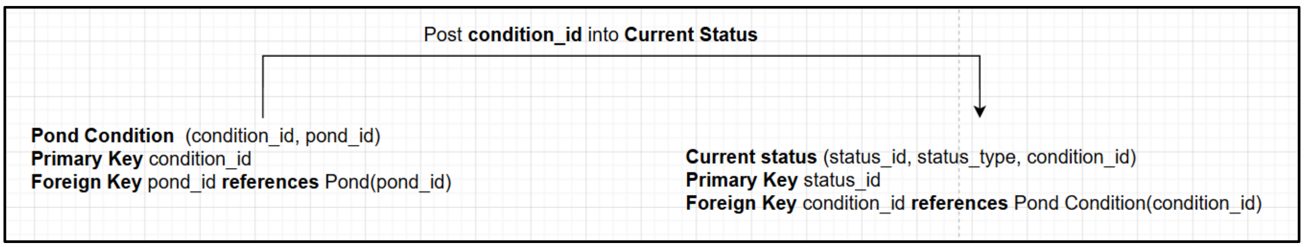
3.5 Complex Relationship Types

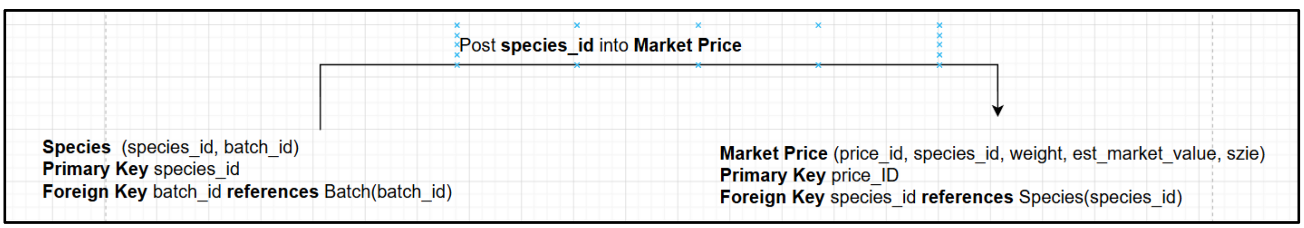




3.6 Multi-Valued Attributes







4.0 Normalization Check

3NF

1) Feeding management (feed_id ,staff_id ,date_feed,time_feed,quantity_food,cost_food)
2) Batch (batch_id, pond_id, StartDate, EndDate, Species)
3) Batch Feeding(feed_id,batch_id)
4) Daily Activity Management (act_id, batch_id, staff_id, actType_id, date_actUsed, time_actUsed, value_actUsed, cost_actUsed)
5) Activity Type (actType_id, act_name,unit_actUsed)
6) Pond (pond_id, pond_loc, pond_size)
7) Water Quality (water_quality_id, pond_id, pH, dissolved_oxygen_lvl), temperature)

8) Health Status (health_condition_id, batch_id, mortality_rates)
9) Pond Condition (condition_id, pond_id)
10) Species (species_id, batch_id)
11) Customer (Customer_id ,Customer_name, Customer_contact, Customer_address)
12) Sales_transaction (Sales_id, Customer_id, Sales_quantity, Price, Transaction_date ,Payment_record)
13) Inventory Item (item_id, item_name, quantity_in_stock, unit, price, expiry_date, location)
14) Supplier (supplier_id, company_name, company_person, company_phone, company_email, address, payment_terms)
15) Purchase Order (po_id, po_date, total_cost, status, payment_terms)
16) PO Detail (po_id, quantity_ordered, unit_cost, item_id)
17) Cost Record (cost_id, cost_type, amount, date, activity_type)
18) Category (category_id, category_name)

5.0 Summary Relationship

Staff (staff_id,staff_name) Primary key staff_id	Pond (pond_id, pond_loc, pond_size) Primary key pond_id
Staff Phone (staff_id,phone_id, ph_num) Primary key phone_id Foreign key staff_id references Staff(staff_id)	Pond Condition (condition_id, pond_id) Primary Key condition_id Foreign Key pond_id references Pond (pond_id)
Staff Role (staff_id, staff_role_id) Primary key staff_id， staff_role_id Foreign key staff_id references Staff (staff_id) Foreign key staff_role_id references Role(staff_role_id)	Current Status (status_id, emptied, cleaned, refilled, reassigned, condition_id) Primary Key status_id Foreign Key condition_id references Pond Condition (condition_id)
Equipment(eq_id,eq_name,eq_type) Primary key eq_id	Role (staff_role_id,role_name) Primary key staff_role_id

Daily Activity Management (act_id, batch_id, staff_id, actType_id, date_actUsed, time_actUsed, value_actUsed, cost_actUsed) Primary key act_id Foreign key actType_id references Activity Type(actType_id) Foreign key batch_id references Batch(batch_id) Foreign key staff_id references Staff(staff_id)	BatchHealthLog (Batch_id, health_condition_id) Primary Key Batch_id, health_condition_id Foreign Key Batch_id references Batch(Batch_id) Foreign Key health_condition_id references Health Status(health_condition_id)
Equipment usage(eqUsage_id, eq_id,act id,eqUsed_timeStart,eqUsed_timeEnd) Primary key eqUsage_id Foreign key eq_id references Equipment(eq_id) Foreign key act_id references Daily Activity Management(act_id)	Health Status (health_condition_id, batch_id, mortality_rates) Primary Key health_condition_id Foreign Key batch_id references Batch (batch_id)
Batch (batch_id, pond_id, StartDate, EndDate, Species) Primary key batch_id Foreign key pond_id references Pond(pond_id)	Treatments (treatment_ID, health_condition_id, treatment_used) Primary Key treatment_ID Foreign Key health_condition_id references Health Status(health_condition_id)
Batch Feeding(feed_id,batch_id) Primary key feed_id,batch_id Foreign key feed_id references Feeding Management(feed_id) Foreign key batch_id references Batch (batch_id)	Species (species_id, batch_id) Primary Key species_id Foreign Key batch_id references Batch(batch_id)
Activity Type(actType_id,act_name, unit_actUsed) Primary key actType_id	Category (category_id, category_name) Primary key category_id

Feeding Management (feed_id, pond_id, staff_id ,date_feed, time_feed,quantity_food,cost_food) Primary key feed_id Foreign key pond_id references Pond(pond_id) Foreign key staff_id references Staff(staff_id)	Sales_transaction (Sales_id, Customer_id, Sales_quantity, Price, Transaction_date ,Payment_record) Primary key Sales_id Foreign key customer_id references Customer(customer_id)
Market Price (price_id, species_id, weight, est_market_value, age, size) Primary key price_id Foreign Key species_id references Species (species_id)	Supplier (supplier_id, company_name, company_person, company_phone, company_email, address, payment_terms) Primary key supplier_id
Purchase Order (po_id, po_date, total_cost, status, payment_terms) Primary key po_id Foreign key supplier_id reference Supplier(supplier_id)	Inventory Item (item_id, item_name, quantity_in_stock, unit, price, expiry_date, location) Primary key item_id Foreign key supplier_id references Supplier(supplier_id) Foreign key category_id references Category(category_id)
PO Detail (po_id, quantity_ordered, unit_cost, item_id) Primary key po_id, item_id Foreign key po_id reference Purchase Order(po_id) Foreign key item_id reference Inventory Item(item_id)	Cost Record (cost_id, cost_type, amount, date, activity_type) Primary key cost_id Foreign key po_id references Purchase Order(po_id) Foreign key batch_id reference Batch(batch_id)

Customer (Customer_id ,Customer_name, Customer_contact, Customer_address) Primary key Customer_id	

6.0 Data Dictionary

Table Name:Staff

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
staff_id	Unique identifier for each staff member.	INT	11	Yes	No	—
staff_name	Full name of the staff member.	VARCHAR	100	No	No	—

Table Name:Staff Phone

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
phone_id	Unique identifier for each phone record.	INT	11	Yes	No	—

staff_id	Link to the specific staff member.	INT	11	No	Yes	Staff
ph_num	The phone number of the staff member.	VARCHAR	15	No	No	—

Table Name:Staff Role

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
staff_id	Unique ID of the staff member assigned to the role.	INT	11	Yes	Yes	Staff
staff_role_id	Unique ID of the specific role being assigned.	INT	11	Yes	Yes	Role

Table Name:Role

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
staff_role_id	Unique identifier for each role type.	INT	11	Yes	No	—

role_name	The descriptive name of the role.	VARCHAR	50	No	No	—

Table Name: Daily Activity Management

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
act_id	Unique identifier for each activity record.	INT	11	Yes	No	—
batch_id	Reference to the specific batch of fish involved.	INT	11	No	Yes	Batch
staff_id	Reference to the staff member performing the task.	INT	11	No	Yes	Staff
actType_id	Reference to the type of activity such as feeding and cleaning.	INT	11	No	Yes	Activity Type

date_actUsed	Date when the activity was performed.	DATE	—	No	No	—
time_actUsed	Time when the activity was performed.	TIME	—	No	No	—
value_actUsed	The amount of something such as chemicals used during an activity.	DECIMAL	10,2	No	No	—
cost_actUsed	The cost spent for that specific activity or usage	DECIMAL	10,2	No	No	—

Table Name: Equipment

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
eq_id	Unique identifier for each piece of equipment.	INT	11	Yes	No	—

eq_name	The name of the equipment such as Water Pump.	VARCHAR	100	No	No	—
eq_type	Category of equipment such as electrical, manual.	VARCHAR	50	No	No	—

Table Name: Equipment usage

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
eqUsage_id	Unique identifier for this specific usage event.	INT	11	Yes	No	—
eq_id	Reference to the equipment being used.	INT	11	No	Yes	Equipment
act_id	Reference to the activity where it was used.	INT	11	No	Yes	Daily Activity Management

eqUsed_timeStart	The exact time the equipment started being used.	TIME	—	No	No	—
eqUsed_timeEnd	The exact time the equipment stopped being used.	TIME	—	No	No	—

Table Name: Activity Type

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
actType_id	Unique identifier for the activity type.	INT	11	Yes	No	—
act_name	Name of the activity such as feeding and Water Change.	VARCHAR	50	No	No	—
unit_actUsed	The standard unit used such as kg, Lilter or hours.	VARCHAR	10	No	No	—

Table Name: Batch Feeding

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
feed_id	Reference to the specific feeding event.	INT	11	Yes	Yes	Feeding Management
batch_id	Reference to the specific batch of fish being fed.	INT	11	Yes	Yes	Batch

Table Name: Batch

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
batch_id	Unique identifier for each fish batch.	INT	11	Yes	No	—
pond_id	Reference to the pond where the batch is located.	INT	11	No	Yes	Pond
StartDate	The date the batch was introduced to the farm.	DATE	—	No	No	—

EndDate	The date the batch was harvested or moved out.	DATE	—	No	No	—
Species	The type of fish in the batch (finger fish.)	VARCHAR	50	No	No	—

Table Name: Feeding Management

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
feed_id	Unique identifier for the feeding event.	INT	11	Yes	No	—
pond_id	Reference to the pond where feeding occurred.	INT	11	No	Yes	Pond
staff_id	Reference to the staff member who fed the fish.	INT	11	No	Yes	Staff
date_feed	The date the feeding took place.	DATE	—	No	No	—
time_feed	The time the feeding took place.	TIME	—	No	No	—

quantity_food	Amount of food used (kg).	DECIMAL	10,2	No	No	—
cost_food	The cost spend for the food used in this session.	DECIMAL	10,2	No	No	—

Table name: Customer

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
Customer_id	Unique identifier for each payment	INT	-	Yes	No	—
Customer_name	Date when payment was made	VARCHAR	100	No	No	—
Customer_address	Payment amount	VARCHAR	255	No	No	—
Customer_contact	Transaction being paid for	VARCHAR	15	No	No	—

Table: Sales_transaction

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
Sales_id	Unique identifier for each sales transaction	INT	-	Yes	No	—

Sales_quantity	Quantity of products sold	DECIMAL	(10, 2)	No	No	—
Price	Price per unit	DECIMAL	(10, 2)	No	No	—
Transaction_date	Date when transaction occurred	DATE	-	No	No	—
Customer_id	Customer who made the purchase	INT	-	No	Yes	Customer

Table Name: Pond

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
pond_id	Unique identifier for each fish pond.	INT	11	Yes	No	—
pond_loc	The physical location where the fish pond is placed.	VARCHAR	11	50	No	—
pond_size	The surface area or volume capacity of the fish pond	DECIMAL	(8,2)	No	No	—

Table Name: Pond Condition

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
condition_id	Unique identifier for the condition of each fish pond.	INT	10	Yes	No	–
pond_id	Reference to the pond to link the condition it is carrying out..	INT	10	No	Yes	Pond

Table Name: Current Status

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
status_id	Unique ID for the specific status update.	INT	10	Yes	No	–
condition_id	Reference to the condition of the pond to link the current state of the pond.	INT	10	No	Yes	Pond Condition
status_type	The type of status that could happen in the pond such as emptied, cleaned, refilled, or reassigned	VARCHAR	10	No	No	–

Table Name: Environmental Monitoring

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
Batch_id	Unique ID to identify the fish batch.	INT	11	Yes	Yes	batch
water_quality_id	Reference to the water quality of the fish pond	INT	11	Yes	Yes	Water Quality

Table Name: Water Quality

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
water_quality_id	Unique identifier for each specific water reading of the fish pond	INT	11	Yes	No	–
pond_id	Reference to the fish pond where the water quality is being tested.	INT	11	No	Yes	Pond
temperature	The measured water temperature in degrees Celsius.	DECIMAL	(3,2)	No	No	–
pH	Parameter to measure how acidic or alkaline the water of the fish pond.	DECIMAL	(3,1)	No	No	–

dissolved _oxygen _lvl	The level of oxygen available in the water (mg/L)	DECIMAL	(5,2)	No	No	–

Table Name: Species

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
species_id	Unique identifier for each fish species farming in this project.	INT	11	Yes	No	–
batch_id	Reference to the fish batch where the fish species belongs to.	INT	11	No	Yes	Batch

Table Name: Market Price

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
price_id	Unique identifier for the prices of each of the species of the fish	INT	10	Yes	No	–
species_id	Reference to the species of fish and how much they cost	INT	11	No	Yes	Species
Weight	The weight threshold for the price (g)	DECIMAL	(6,2)	No	No	–

est_market_value	The current market price for this specific size and weight	DECIMAL	(8,2)	No	No	–
Size	The size category for this price (e.g., “Small”, “Medium”, “Large”)	VARCHAR	20	No	No	–

Table Name: BatchHealthLog

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
Batch_id	Unique identifier for the batch for Health log	INT	11	Yes	Yes	Batch
health_condition_id	Reference to the health condition of the fish based on each batch.	INT	11	Yes	Yes	Health Status

Table Name: Health Status

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
Health_status_id	Unique identifier for the specific health check session	INT	11	Yes	No	–

batch_id	Reference to the fish batch of their health status.	INT	11	No	Yes	Batch
Mortality rates	The percentage of fish loss found during the check.	DECIMAL	(5,2)	No	No	–

Table Name: Supplier

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
supplier_id	Unique identifier assigned to each supplier	INT	5	Yes	No	-
company_name	Registered name of the supplier company	VARCHAR	50	No	No	-
company_person	Name of the contact person representing the supplier	VARCHAR	50	No	No	-
company_phone	Contact phone number of the supplier	VARCHAR	15	No	No	-
company_email	Official email address of the supplier	VARCHAR	50	No	No	-
address	Business address of the supplier	VARCHAR	100	No	No	-
payment_terms	Agreed payment terms with the supplier (e.g. credit period)	VARCHAR	30	No	No	-

Table Name: Category

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
category_id	Unique identifier for each item category	INT	5	Yes	No	-
category_name	Name used to classify inventory items	VARCHAR	30	No	No	-

Table Name: Inventory Item

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
item_id	Unique identifier for each inventory item	INT	5	Yes	No	-
item_name	Name or description of the inventory item	VARCHAR	50	No	No	-
quantity_in_stock	Current quantity of the item available in inventory	INT	6	No	No	-
unit	Unit of measurement for the item (e.g. kg, pcs)	VARCHAR	10	No	No	-
price	Unit price of the inventory item	DECIMAL	8,2	No	No	-
expiry_date	Expiration date of the item (if applicable)	DATE	-	No	No	-
location	Storage location of the item within the facility	VARCHAR	50	No	No	-

supplier_id	Identifier of the supplier providing the item	INT	5	No	Yes	Supplier
category_id	Identifier of the category the item belongs to	INT	5	No	Yes	Category

Table Name: Purchase Order

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
po_id	Unique identifier for each purchase order	INT	5	Yes	No	-
po_date	Date when the purchase order is issued	DATE	-	No	No	-
total_cost	Total cost of all items included in the purchase order	DECIMAL	10,2	No	No	-
status	Current status of the purchase order (e.g. pending, completed)	VARCHA R	20	No	No	-
payment_t erms	Payment conditions agreed for the purchase order	VARCHA R	30	No	No	-
supplier_i d	Identifier of the supplier fulfilling the order	INT	5	No	Yes	Supplier

Table Name: PO Detail

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
po_id	Identifier of the related purchase order	INT	5	Yes	Yes	Purchase Order
item_id	Identifier of the inventory item in order	INT	5	Yes	Yes	Inventory Item
quantity_ordered	Quantity of the item ordered in the purchase order	INT	6	No	No	-
unit_cost	Cost per unit of the item at the time of ordering	DECIMAL	8,2	No	No	-

Table Name: Cost Record

Column Name	Description	Data Type	Size	Primary key?	Foreign Key?	FK referenced table
cost_id	Unique identifier for each cost record	INT	5	Yes	No	-
cost_type	Category or type of cost incurred (e.g. transport, handling)	VARCHAR	30	No	No	-
amount	Monetary value of the cost recorded	DECIMAL	10,2	No	No	-
date	Date when the cost was recorded	DATE	-	No	No	-

activity_type	Type of activity associated with the cost	VARCHAR	30	No	No	-
po_id	Identifier of the related purchase order	INT	5	No	Yes	Purchase Order
batch_id	Identifier of the related batch for the cost	INT	-	No	Yes	Batch

7.0 Logical ER Diagram